ACTIVITY 4: Testing the Enhanced Account Class

Objective

Write a test class that verifies all the new features using boolean return values.

Requirements

Create a test class `TestAccountEnhanced.java` that tests:

1. Valid account creation
2. Invalid age (under 18) - should auto-correct to 18
3. Invalid account type - should default to "Savings"
4. Minimum balance enforcement on creation - should auto-correct
5. Minimum balance enforcement on withdrawal - should return false
6. Account status management (close and reopen)
7. PIN protection (set, verify, correct/incorrect PIN)

Expected Output

When you run your test class, the output should look like this:

text

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  ENHANCED ACCOUNT TEST (BOOLEAN RETURNS)
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>>> Test 1: Valid Account Creation
Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No

>>> Test 2: Invalid Age (under 18)
Creating account with age 16
Age auto-corrected to: 18
Account #1002 | Young Kid (18 yrs) | Savings | ₹500.0 | Active | PIN: No

>>> Test 3: Invalid Account Type
Creating account with type "Invalid"
Account type defaulted to: Savings
Account #1003 | Test User (25 yrs) | Savings | ₹500.0 | Active | PIN: No

>>> Test 4: Minimum Balance Enforcement on Creation
Creating Savings account with ₹300 (below minimum)
Balance auto-corrected to minimum: ₹500.0
Account #1004 | Bob Wilson (25 yrs) | Savings | ₹500.0 | Active | PIN: No

>>> Test 5: Withdrawal with Minimum Balance
Initial: Account #1005 | Alice Brown (30 yrs) | Current | ₹1000.0 | Active | PIN: Yes
Withdrawing ₹200.0: SUCCESS
New balance: ₹800.0
After withdrawal: Account #1005 | Alice Brown (30 yrs) | Current | ₹800.0 | Active | PIN: Yes
Withdrawing ₹900.0 (would leave ₹-100): FAILED (Minimum balance violation)
Current balance: ₹800.0

>>> Test 6: Account Status Management
Initial: Account #1006 | Charlie Green (35 yrs) | Savings | ₹2000.0 | Active | PIN: No
Closing account: SUCCESS
After close: Account #1006 | Charlie Green (35 yrs) | Savings | ₹2000.0 | Inactive | PIN: No
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Depositing ₹500.0 to closed account: FAILED (Account inactive)
Reopening account: SUCCESS
After reopen: Account #1006 | Charlie Green (35 yrs) | Savings | ₹2000.0 | Active | PIN: No

>>> Test 7: PIN Protection
Setting PIN 1234: SUCCESS
Withdrawing ₹200.0 with correct PIN (1234): SUCCESS
New balance: ₹1300.0
Withdrawing ₹100.0 with incorrect PIN (9999): FAILED (Incorrect PIN)
Withdrawing ₹100.0 with PIN not set: FAILED (PIN not set)

>>> Test 8: All Accounts Summary
Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No
Account #1002 | Young Kid (18 yrs) | Savings | ₹500.0 | Active | PIN: No
Account #1003 | Test User (25 yrs) | Savings | ₹500.0 | Active | PIN: No
Account #1004 | Bob Wilson (25 yrs) | Savings | ₹500.0 | Active | PIN: No
Account #1005 | Alice Brown (30 yrs) | Current | ₹800.0 | Active | PIN: Yes
Account #1006 | Charlie Green (35 yrs) | Savings | ₹2000.0 | Active | PIN: No
Account #1007 | Diana Prince (28 yrs) | Savings | ₹1300.0 | Active | PIN: Yes

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  ENHANCED TEST COMPLETED!
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Task

Write the complete `TestAccountEnhanced` class in a file called `TestAccountEnhanced.java`. Handle all display formatting in this test class.